

Final MS Word Report

This study's purpose was to analyze the relationship between smoking status and body mass index with data from the National Health Interview Survey (NHIS) among adults in the United States. Data from the National Health Interview Survey is relevant to different aspects of health collected and surveyed by the Center for Disease Control and Prevention. Having the understanding of the potential relationship between smoking and BMI is crucial because of how prevalent the health issues of smoking and obesity are in public health and the chronic illnesses that may come from it. The research question at hand is: Does the BMI in adults across the United States have significance in smoking status groups? The null hypothesis is there is no significant difference between BMI and smoking status groups. The alternative hypothesis would be that there is a significance between BMI and smoking status groups.

The dataset provided by NHIS contained both categorical and quantitative variables. The variables include age, BMI, income, sex, smoking status, education level, and urban or rural residence. To begin analyzing the data through RStudio, I have cleaned the data and created categories. Missing values were cleaned from the dataset to only include valid data. The numeric value for sex was changed to Male and Female, smoking status was categorized into current smoker, former smoker, and never smoke, and age was grouped into relative age groups.

The utilization of descriptive statistics helped me summarize the quantitative variables. Measures of central tendencies were utilized to analyze age and body mass index. Moreover, frequency analyses were utilized to analyze categorical variables such as sex, smoking status, and education level. From the frequency analyses, we find that the majority of the participants were female and

most respondents are former smokers. There was a smaller occurrence of current smokers. When analyzing education levels, we found that most participants' education has consisted of high school. These descriptive statistics showed the study sample's demographics.

To further analyze the data and test the hypothesis, an ANOVA test was implemented. The results of the ANOVA test showed the F-statistic of 0.691 and a p-value of 0.406. With this data, we can conclude that the hypothesis is null and could not be rejected; the results were not statistically significant. This translated to the conclusion that there is no significant difference between BMI and smoking status groups in adults across the US.

With the results of the analysis of the NHIS dataset from RStudio being not statistically significant, there was a magnified limitation in the study. The limitation of the study was the number of invalid observations and the high rate of missing data.

Therefore, utilizing RStudio to produce an analysis on the relationship between BMI and smoking status groups among adults in the US has led to the conclusion that there is no significant difference between BMI and smoking status groups in adults across the US. The evaluation was confirmed amongst the usage of ANOVA testing, multivariate visualizations, and descriptive statistics. If this study were to be performed again, I would encourage researchers to get a larger sample size to account for missing data and to also account for more variables that could affect lifestyles to encourage smoking status.